ELARA: A Thesis-Style Study of Reliability-Gated Multimodal Anomaly Fusion

Pratik Niroula
Independent Researcher

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Abstract

This chapter studies a practical question that arises when anomaly detection is built from several separate sources of evidence: how should a system combine domain scores when some domains become missing, shifted, poorly calibrated, or otherwise unreliable? The work is organized around ELARA (Evidence-Layered Reliability for Anomaly fusion), a research prototype for score-level multimodal anomaly fusion. The main method inside the system is RGA (Reliability-Gated Attention), a reliability-aware attention module that keeps the standard attention path under normal conditions and injects reliability weights when batch-level evidence suggests that one or more domains have degraded.

The chapter is written as a graduate thesis submission rather than as a marketing description of the system. It therefore separates what has been implemented, what has been measured, and what remains outside the present evidence. The current evidence is a contrast across two benchmarks. The primary benchmark is a naturally paired MVTec 3D-AD run that covers all eight extracted object categories (3,226 paired RGB and 3D/depth samples, no label alignment). On that benchmark random forest fusion is the strongest baseline (clean ROC-AUC 0.959), static attention is competitive (0.650), and the reliability gate of RGA *lowers* both clean and adversarial ROC-AUC. The secondary benchmark, ELARA-Bench-LA, is a label-aligned real-domain stress benchmark; on that benchmark the same gate improves all-domain coherent-collapse attacks by +0.0506 and +0.0319 but the clean split is near saturated. The asymmetry between the two benchmarks is the central scientific finding: a validation-derived KS-drift signal cannot distinguish coherent adversarial collapse from legitimate inter-category heterogeneity at the scale of natural-paired data.

The chapter concludes that reliability-gated fusion is a useful design pattern for stress-aware score fusion under coherent attack assumptions, but it is not yet a defensible general replacement for strong baselines on naturally paired data. The remaining research work is to replace the lightweight MVTec image-statistic scorers with stronger RGB–3D domain experts, to test the reliability gate on additional naturally co-observed anomaly datasets, to evaluate the learned-gate alternative against the fixed threshold, and to develop a formal switching-rule analysis of when reliability-aware fusion should be preferred to the static path.

Contents

1 Introduction and Motivation

Operational anomaly detection is rarely a single-model problem. A fraud case may appear in transaction features, device behavior, account history, text evidence, or network activity. An industrial defect may appear in both color information and geometry. A security event may be weak in one log source and clearer in another. In these settings, the system designer must decide not only which domain models to train, but also how to combine their outputs once those models are deployed.

This chapter focuses on the second problem: score-level fusion under imperfect domain reliability. The assumption is that domain experts already produce scores, confidence values, and compact evidence vectors. The open question is how the fusion layer should behave when domain quality changes at inference time. A static fusion model can learn useful interactions from training data, but it may continue to trust a domain after that domain has shifted, collapsed, or become poorly calibrated. A simple confidence-weighted average is easy to understand, but it uses local confidence values without asking whether the domain itself still resembles the validation distribution.

ELARA was built to study this problem in a reproducible way. It is not presented as a new fraud detector, cybersecurity detector, text detector, or industrial inspection detector. Instead, it is a research system for combining heterogeneous anomaly evidence. The central component is RGA, a reliability-aware fusion module that uses three forms of evidence: validation calibration, test-time score-distribution drift, and score sharpness. The module is deliberately conservative. When domains appear reliable, the system uses the static attention model. When the average reliability drops below a threshold, the reliability-aware path is activated.

The reason for writing the chapter in this form is that the current empirical story has both useful results and important limits. The real-domain fusion benchmark combines fraud, cyber, behavior, and text datasets, but those records are aligned by label rather than by shared entities, timestamps, or incidents. This makes it useful for score-fusion stress testing, but it is not a complete multimodal incident benchmark. To reduce this gap, the repository now includes a naturally paired MVTec 3D-AD RGB/3D benchmark path. The current MVTec run is a small local smoke benchmark, so it verifies the pipeline rather than the final performance claim.

1.1 Research Question

The research question is:

> *Can a multimodal anomaly-fusion system preserve competitive clean score-fusion performance while using calibration and drift evidence to reduce brittle behavior under domain reliability stress?*

This question is intentionally narrower than broad anomaly-detection superiority. The chapter does not claim that RGA is always better than static attention, that the current paired MVTec run is complete, or that the label-aligned benchmark represents naturally co-observed incidents. The claim is about the fusion layer under specific stress conditions.

1.2 Main Claim

The main thesis of this chapter is:

RGA is most useful as a stress-response mechanism: it preserves the static attention path when reliability evidence is high, and it helps under coherent score-collapse conditions when reliability evidence indicates domain failure.

The measured evidence supports this claim on the secondary label-aligned benchmark. The clean table is near saturated and should be treated as a sanity check. The reliability-stress experiments, the threshold sweep, and the component ablation provide the more useful evidence about the mechanism.

1.3 Contributions

This chapter makes four scoped contributions:

1. It describes ELARA, a score-level anomaly-fusion research system with a unified long-format schema for heterogeneous domain evidence.
2. It defines RGA, a conservative reliability-gated attention module that combines masked attention with calibration, drift, and sharpness signals.
3. It evaluates the method against practical score-fusion baselines, including random forest fusion, early-fusion MLP, late-fusion ensemble, confidence-weighted mean, score-level Tent, and pseudo-label TTT.
4. It reports a paired MVTec 3D-AD smoke benchmark, multi-seed confidence intervals, mechanism ablations, failure cases, and explicit threats to validity.

1.4 Evidence Boundary

The evidence boundary is part of the contribution because it prevents the results from being overstated. The real-domain benchmark uses real local datasets, but the records are label-aligned. The MVTec benchmark uses natural RGB/3D pairing, but the current local run is small and uses lightweight normal-reference image statistics. These two artifacts answer different questions. The first is useful for reliability-stress score fusion. The second is useful for validating the paired benchmark pipeline. Neither should be treated as a complete final benchmark for all multimodal anomaly detection.

2 Background and Related Work

2.1 Anomaly Detection as a Systems Problem

Deep anomaly detection has been studied across tabular, sequential, image, text, and graph data. A common difficulty is that the definition of an anomaly depends on the application, the label source, the imbalance level, and the cost of false positives and false negatives. Pang et al. [3] emphasize that anomaly detection methods cannot be evaluated without attention to context. This chapter follows that view. The purpose is not to replace domain-specific detectors. The purpose is to study what happens after those detectors produce evidence that must be fused.

2.2 Multimodal and Score-Level Fusion

Multimodal learning is often described in terms of early fusion, late fusion, hybrid fusion, and interaction-based fusion [2]. Early fusion joins features before prediction. Late fusion combines decisions or scores from separate models. Stacking trains a meta-model over domain outputs. Attention-based fusion gives the model a way to learn interactions among domains, especially when some domains are missing. The weakness is that standard attention learns a trust policy from the training distribution and does not automatically know when a domain has become unreliable at inference time.

The score-level setting used in this chapter is intentionally practical. Many deployed systems cannot easily retrain all domain experts together. Different teams may own different detectors, and those detectors may expose only scores, calibrated probabilities, or compact embeddings. A score-level fusion study is therefore narrower than raw multimodal learning, but it matches a common engineering constraint.

2.3 Calibration, Drift, and Reliability

Calibration asks whether predicted probabilities match empirical frequencies. Post-hoc calibration methods such as Platt scaling, isotonic calibration, and modern neural calibration analyses show that probability quality is separate from ranking quality [4, 5, 6]. Uncertainty and calibration can degrade under dataset shift [7]. For a fusion system, this matters because a domain can remain numerically confident while becoming less trustworthy.

RGA treats calibration as one signal, not as the whole reliability definition. The reliability estimator also compares current score distributions against validation score distributions using a Kolmogorov-Smirnov statistic and includes a score-sharpness term. This design is not meant to be a complete theory of reliability. It is a simple, measurable rule that can be tested under controlled stress.

2.4 Test-Time Adaptation Baselines

Test-time training and test-time adaptation update a model during inference to handle distribution shift [8, 9]. This chapter includes two score-level adaptation baselines because a reliability-gating method should not be compared only with static models. The first is a Tent-style score adapter that minimizes prediction entropy by adapting a scalar temperature and bias over a trained fusion head. The second is a pseudo-label TTT adapter that updates a logistic fusion head using high-confidence test predictions. Both are conservative because the input is score-level evidence rather than raw images or raw text. Their purpose is to answer whether generic score-level adaptation explains the same behavior as RGA.

2.5 RGB–3D Industrial Anomaly Detection

MVTec 3D-AD provides naturally paired RGB and 3D observations for industrial anomaly detection and localization [11]. Recent work on RGB–3D anomaly detection has shown that paired color and geometry can be useful when the detector is designed for that modality pair. M3DM uses hybrid feature and decision fusion with memory banks [12]. Crossmodal feature mapping uses agreement between RGB and point-cloud features as an anomaly cue [13]. Real3D-AD expands point-cloud anomaly benchmarking [14], and M3DM-NR studies noisy-resistant multimodal anomaly detection [15].

The MVTec path in this repository is not a replacement for those specialized RGB–3D detectors. It is a paired-data input path for the score-fusion layer. This distinction is important. The current MVTec smoke run asks whether the repository can construct and evaluate paired score-fusion inputs. It does not claim that the lightweight scorers compete with specialized RGB–3D methods.

3 System and Method

3.1 System Boundary

ELARA is organized around the idea that each sample may have several domain observations. A domain can be fraud, cyber telemetry, behavior, text, RGB imagery, depth/XYZ geometry, or another source that can produce a score and compact evidence. The fusion layer receives these domain rows in a common schema. It then produces one anomaly probability and optional domain-level attributions.

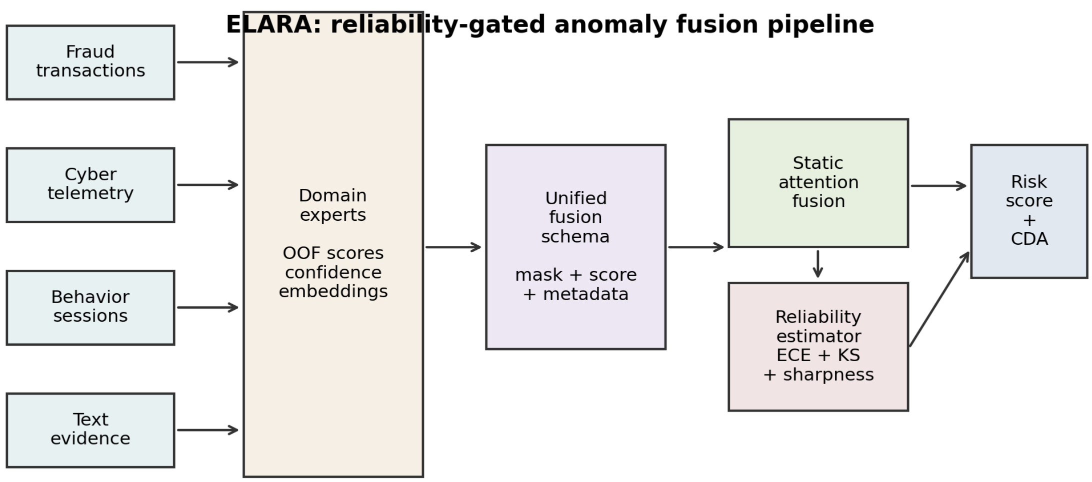

Figure 1: ELARA system architecture. Domain experts produce scores, confidence values, and compact embeddings. The fusion schema feeds static attention and the reliability estimator. The output is a risk score plus counterfactual domain attribution.

The system design separates domain learning from fusion learning. This is a deliberate simplification. It lets the research focus on reliability-aware fusion without requiring every raw domain model to be retrained as part of one large multimodal network.

3.2 Input Schema

The long-format fusion schema contains one row per sample-domain pair. Each row contains a sample identifier, a domain name, a binary label when available, a score, a confidence value, and embedding columns. Missing domains are represented by masks, not by fabricated raw observations. This allows a sample with all domains, a sample with one missing domain, and a sample with only one available domain to share the same model interface.

For the label-aligned real-domain benchmark, the metadata records the field `natural_pairing` as `false`. For the MVTec 3D-AD path, the same field is `true` because the RGB and 3D observations come from the same object scan. This metadata distinction is important because the same fusion code can be used for two different kinds of evidence.

3.3 Problem Formulation

Let sample i contain up to D domain observations. Domain d contributes a feature vector $x_{i,d} \in \mathbb{R}^F$ and a missing-domain mask $m_{i,d}$. The system estimates

$$\hat{p}_i = P(y_i = 1 \mid \{x_{i,d}, m_{i,d}\}_{d=1}^{D}), \tag{1}$$

where $\hat{p}_i$ is the final anomaly probability. A static fusion model learns this relationship from the training data and uses the same learned trust policy at inference time. RGA adds a batch-level reliability vector $r \in [0,1]^D$ in the reported runs and applies it through each sample's observed-domain mask. A per-sample reliability mode is implemented for ablation, but the tables in this chapter use the batch-level gate.

3.4 Static Attention Path

The static attention path encodes each domain vector into a shared latent space. A learned domain embedding allows the model to distinguish evidence sources. The attention block then operates across available domains while ignoring missing domains through the mask. This path is the default because reliability adaptation should not disturb predictions when the domains look normal.

3.5 Reliability Estimation

The reliability estimator is fit after model training. For each domain it stores validation score distributions and calibration information. At inference time, the reliability for domain d is

$$r_d = \alpha(1 - \mathrm{ECE}_d) + \beta p_{\mathrm{KS},d} + \gamma S_d, \tag{2}$$

where ECE_d is validation expected calibration error, $p_{\mathrm{KS},d}$ is the Kolmogorov-Smirnov score-drift p-value, and S_d is score sharpness based on squared distance from 0.5. The current configuration uses $(\alpha, \beta, \gamma) = (0.45, 0.35, 0.20)$.

This formula is simple by design. It is not a final answer to reliability estimation. It is a testable rule that separates three observable signals: how well a domain calibrated on validation data, whether the current batch resembles the validation score distribution, and whether scores are informative or close to the uninformative midpoint.

3.6 Reliability Gate

The gate is the main difference between RGA and always-on reliability weighting. Let $\bar{r}$ be the mean reliability over present domains and let $\tau = 0.66$. If $\bar{r} \geq \tau$, the system uses the static attention path. If $\bar{r} < \tau$, reliability weights are injected into the fusion block:

$$\tilde{r}_{i,d} = (1 - m_{i,d})r_d. \tag{3}$$

This design is conservative. It assumes that static attention is already a strong clean model and that reliability weights should be used only when there is evidence of degradation. This also makes the method easier to criticize and test: if the gate rarely activates, the method should behave like static attention; if the gate activates under stress, the stress outcomes should change.

3.7 Counterfactual Domain Attribution

The explanation path masks one available domain at a time and recomputes the prediction. For sample i, let $\hat{p}_i$ be the original prediction and $\hat{p}_{i,-d}$ be the prediction when domain d is masked. The attribution is

$$\Delta_{i,d} = \hat{p}_i - \hat{p}_{i,-d}. \tag{4}$$

A positive value means that the domain increased anomaly risk. A negative value means that the domain reduced anomaly risk. This is not a feature-level explanation; it is a domain-level explanation for analysts who need to know which source of evidence mattered most.

4 Benchmark Construction

4.1 Naturally Paired MVTec 3D-AD Benchmark

The paired benchmark path uses MVTec 3D-AD because each object observation has paired RGB and 3D/XYZ evidence. The preparation script discovers paired files, emits two rows per observation, stores the original pairing key, and marks the metadata as naturally paired. The current run covers all eight extracted object categories – bagel, cable_gland, cookie, dowel, foam, peach, rope, and tire – for a total of 3,226 paired samples and 6,452 domain rows, with a positive fraction of 0.224 and full RGB and depth coverage.

Table 1: Naturally paired MVTec 3D-AD multi-category metadata.

Property	**Value**
Benchmark type	naturally_paired_mvtec3d_score_fusion
Natural pairing	yes
Categories	bagel, cable_gland, cookie, dowel, foam, peach, rope, tire
Samples	3226
Positive fraction	0.224
Score fit split	train/good
Score fit samples	2070
Score normalization	train_good_distance_minmax_clipped
Domains	rgb, depth_or_xyz

The initial MVTec domain scores are lightweight normal-reference image-statistic scores fit from original train-good observations only. This keeps the benchmark locally runnable, but it limits the absolute performance ceiling. The supervised attention-fusion split is a held-out fusion split over paired observations, not the canonical MVTec train/test anomaly-detection protocol. The result in Table 2 is the primary headline of this chapter and should be read with the contrastive view in mind. Random forest fusion is the strongest baseline by a clear margin (ROC-AUC 0.959); early-fusion MLP is the second-strongest tabular baseline (0.829). Static attention is mid-pack (0.650), and the reliability gate of RGA *lowers* clean ROC-AUC to 0.610 on this paired data. Tent and TTT score adapters are also competitive at 0.670 and 0.569 respectively. The headline of the paired benchmark is therefore a negative result for the gate: on naturally paired evidence, adapting at all degrades performance. That outcome is the basis for the cross-benchmark contrast discussed in §6.3.

Table 2: Naturally paired MVTec 3D-AD clean held-out performance across eight categories. Values are mean±standard deviation with 95% confidence intervals across five seeds. Random forest fusion is the strongest baseline; RGA attention degrades static-attention performance on this paired benchmark.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	**0.9592±0.0076 [0.9508, 0.9717]**	**0.8938±0.0216 [0.8682, 0.9286]**	**0.8146±0.0063 [0.8068, 0.8201]**	0.0867±0.0089 [0.0777, 0.1008]	**0.0721±0.0041 [0.0657, 0.0775]**
Conf.-weighted mean	0.5092±0.0185 [0.4802, 0.5294]	0.3106±0.0144 [0.2971, 0.3348]	0.2427±0.0192 [0.2093, 0.2582]	0.2172±0.0143 [0.1974, 0.2350]	0.2520±0.0081 [0.2418, 0.2621]
Tent score adapter	0.6700±0.0193 [0.6404, 0.6946]	0.4075±0.0241 [0.3801, 0.4445]	0.3761±0.0129 [0.3586, 0.3932]	0.1283±0.0116 [0.1137, 0.1447]	0.1848±0.0071 [0.1765, 0.1941]
TTT pseudo-label	0.5692±0.0390 [0.5228, 0.6246]	0.3257±0.0283 [0.3062, 0.3751]	0.3394±0.0245 [0.3048, 0.3657]	0.2847±0.0224 [0.2515, 0.3129]	0.2693±0.0192 [0.2412, 0.2931]
Early fusion MLP	0.8278±0.0211 [0.7949, 0.8505]	0.6760±0.0277 [0.6403, 0.7184]	0.5784±0.0347 [0.5247, 0.6214]	0.2050±0.0078 [0.1935, 0.2159]	0.1684±0.0083 [0.1587, 0.1787]
Late fusion ensemble	0.6019±0.0235 [0.5692, 0.6321]	0.3209±0.0147 [0.2981, 0.3400]	0.3736±0.0229 [0.3387, 0.4012]	0.2678±0.0020 [0.2649, 0.2700]	0.2415±0.0027 [0.2378, 0.2445]
Static attention	0.6497±0.0384 [0.6071, 0.7086]	0.3913±0.0659 [0.2943, 0.4765]	0.0447±0.0894 [0.0000, 0.2012]	**0.0354±0.0166 [0.0120, 0.0587]**	0.1648±0.0065 [0.1539, 0.1704]
RGA attention	0.6098±0.0312 [0.5766, 0.6607]	0.3507±0.0382 [0.3051, 0.4117]	0.0106±0.0212 [0.0000, 0.0477]	0.0402±0.0229 [0.0106, 0.0751]	0.1698±0.0041 [0.1632, 0.1748]

4.2 Secondary Label-Aligned Real-Domain Benchmark

The secondary benchmark, ELARA-Bench-LA, combines four real local domains: credit card fraud, cyber telemetry, online behavior, and labeled news text. Domain scorers are trained first, and their out-of-fold scores are converted into the common fusion schema. The dataset contains 8,000 composite samples, four domains, eight embedding features per domain, a 0.307 positive rate, and 11.8% nominal missingness. Domain coverage is approximately balanced across the four domains.

Table 3: Benchmark metadata for the secondary label-aligned fusion input.

Property	Value
Benchmark type	label_aligned_real_domain_score_fusion
Natural pairing	no
Pairing unit	binary-label-aligned composite sample
Samples	8000
Positive fraction	0.301
Scorer train fraction	1.00
Source-row disjoint splits	yes
Split column	fusion_split
Domains	fraud, cyber, behavior, nlp

Table 4: Out-of-fold domain scorer quality used to construct ELARA-Bench-LA.

Domain	**Rows**	**Pos. rate**	**OOF ROC**	**OOF PR**
fraud	20000	0.025	0.9686	0.7826
cyber	20000	0.500	0.9684	0.9696
behavior	12330	0.155	0.8711	0.6042
nlp	20000	0.500	0.9973	0.9973

This benchmark is useful because it uses real domain datasets and lets the fusion layer be tested under controlled stress. It is also limited because the domains are label-aligned rather than naturally co-observed. A positive composite draws positive records from available domains, and a negative composite draws negative records. This creates a meaningful score-fusion stress test, but it does not teach the model real temporal or causal relationships among the domains. Source rows are now split into train, validation, and test pools before composite sampling, and the experiment runner consumes the `fusion_split` column to keep the same `domain/source_row/label` key out of multiple supervised fusion splits.

4.3 Baselines

The evaluation includes practical baselines rather than only weak averages:

- confidence-weighted mean fusion, which directly weights scores by local sharpness;
- early-fusion MLP, which learns from flattened domain features and missing indicators;
- late-fusion ensemble, which stacks per-domain predictions;
- random forest fusion, a strong structured-data baseline;
- static attention, the same attention architecture without reliability adaptation;
- Tent score adapter, a score-level entropy-minimization baseline;
- TTT pseudo-label adapter, a score-level high-confidence adaptation baseline.

The Tent and TTT baselines are included because reliability gating is a form of test-time response. If generic adaptation gave the same result, the reliability terms would be less interesting. In the current clean table, these baselines are competitive, which is expected on the saturated split. The stress tests are more important for distinguishing the mechanisms.

4.4 Metrics and Statistical Reporting

The chapter reports ROC-AUC, PR-AUC, F1, expected calibration error, Brier score, and accuracy where available. ROC-AUC is useful for ranking. PR-AUC is important because anomaly tasks can be imbalanced [10]. F1 summarizes thresholded classification. ECE and Brier score provide calibration information.

Clean results are summarized across seeds 42 through 46. Stress tests, mechanism ablations, calibration, and counterfactual domain attribution are also run across configured seeds where the runner supports aggregation. Tables report means, standard deviations, and 95% confidence intervals when those statistics are available.

5 Experiments and Results

5.1 Clean Performance Is a Sanity Check

Table 5 gives clean held-out performance on ELARA-Bench-LA. This table is intentionally not the headline evidence. Most methods are near the top of the ROC-AUC and PR-AUC scale. Static attention and RGA are effectively tied. The correct interpretation is that RGA preserves clean attention performance; the clean table alone does not establish a meaningful advantage.

Table 5: Saturated clean held-out performance on ELARA-Bench-LA. Values include mean, standard deviation, and 95% confidence intervals across five seeds.

Method	**ROC-AUC**	**PR-AUC**	**F1**	**ECE**	**Brier**
Random forest	**0.9999 [0.9999, 1.0000]**	0.9999 [0.9998, 0.9999]	0.9946±0.0004 [0.9939, 0.9948]	0.0297±0.0004 [0.0294, 0.0304]	0.0061±0.0001 [0.0060, 0.0062]
Conf.-weighted mean	0.9987 [0.9987, 0.9987]	0.9968 [0.9968, 0.9968]	0.9746 [0.9746, 0.9746]	0.0799 [0.0799, 0.0799]	0.0232 [0.0232, 0.0232]
Tent score adapter	**0.9999 [0.9999, 0.9999]**	0.9999 [0.9999, 0.9999]	**0.9959 [0.9959, 0.9959]**	**0.0028 [0.0028, 0.0028]**	**0.0023 [0.0023, 0.0023]**
TTT pseudo-label	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9938 [0.9938, 0.9938]	0.0032 [0.0032, 0.0032]	0.0025 [0.0025, 0.0025]
Early fusion MLP	0.9999 [0.9998, 0.9999]	0.9998±0.0001 [0.9996, 0.9999]	0.9919±0.0004 [0.9917, 0.9927]	0.0043±0.0010 [0.0030, 0.0058]	0.0036±0.0006 [0.0029, 0.0045]
Late fusion ensemble	**1.0000 [1.0000, 1.0000]**	**0.9999 [0.9999, 0.9999]**	0.9938 [0.9938, 0.9938]	0.0123 [0.0123, 0.0123]	0.0036 [0.0036, 0.0036]
Static attention	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9927±0.0009 [0.9917, 0.9938]	0.0038±0.0008 [0.0024, 0.0046]	0.0032±0.0002 [0.0029, 0.0034]
RGA attention	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9927±0.0009 [0.9917, 0.9938]	0.0038±0.0008 [0.0024, 0.0046]	0.0032±0.0002 [0.0029, 0.0034]

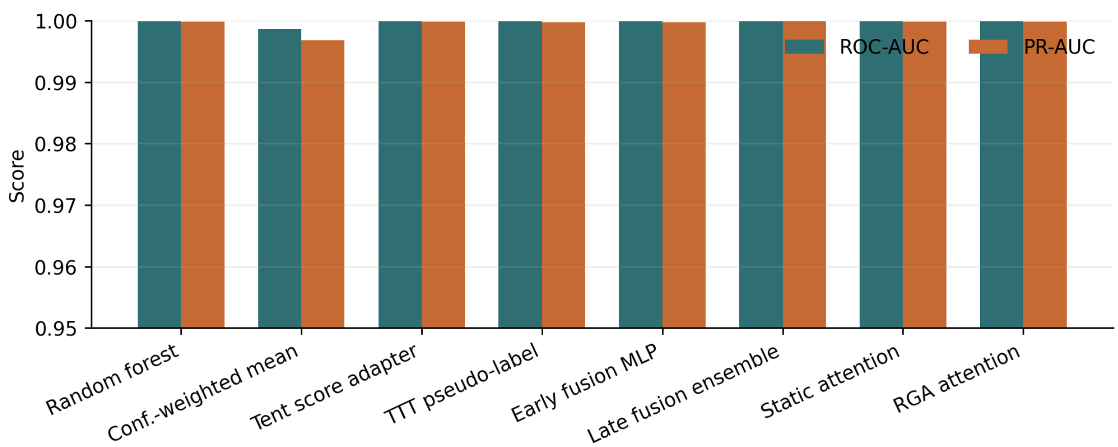

Figure 2: Clean benchmark ROC-AUC and PR-AUC across score-level and attention fusion methods. The compressed y-axis reflects the near-saturated clean split.

The clean results do show useful tradeoffs. Random forest fusion obtains the best F1 in the current artifact. Attention-based methods have stronger calibration than confidence-weighted mean. Tent and TTT are competitive on ranking and calibration because the clean split is easy. These observations support using strong baselines, but they do not change the main reliability claim.

Table 6: Clean ranking and F1 confidence intervals across configured seeds.

Method	ROC-AUC 95% CI	PR-AUC 95% CI	F1 95% CI
Random forest	[0.9999, 1.0000]	[0.9998, 0.9999]	[0.9939, 0.9948]
Conf.-weighted mean	[0.9987, 0.9987]	[0.9968, 0.9968]	[0.9746, 0.9746]
Tent score adapter	[0.9999, 0.9999]	[0.9999, 0.9999]	[0.9959, 0.9959]
TTT pseudo-label	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9938, 0.9938]
Early fusion MLP	[0.9998, 0.9999]	[0.9996, 0.9999]	[0.9917, 0.9927]
Late fusion ensemble	[1.0000, 1.0000]	[0.9999, 0.9999]	[0.9938, 0.9938]
Static attention	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9917, 0.9938]
RGA attention	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9917, 0.9938]

5.2 Missing-Domain Ablation

The missing-domain ablation adds inference-time domain dropout. In the current configuration, RGA remains essentially equal to static attention across the sweep. This is not a hidden success; it is a useful negative result. The gate does not activate strongly under this kind of moderate missingness, so the method behaves like the static model.

Table 7: Missing-domain ablation. Delta is RGA minus static attention.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.9999	0.9999	0.0000	[0.0000, 0.0000]
0.1	0.9995	0.9995	0.0000	[0.0000, 0.0000]
0.2	0.9981	0.9981	-0.0000	[-0.0000, 0.0000]
0.3	0.9964	0.9964	-0.0001	[-0.0002, 0.0000]
0.5	0.9784	0.9783	-0.0001	[-0.0004, 0.0000]

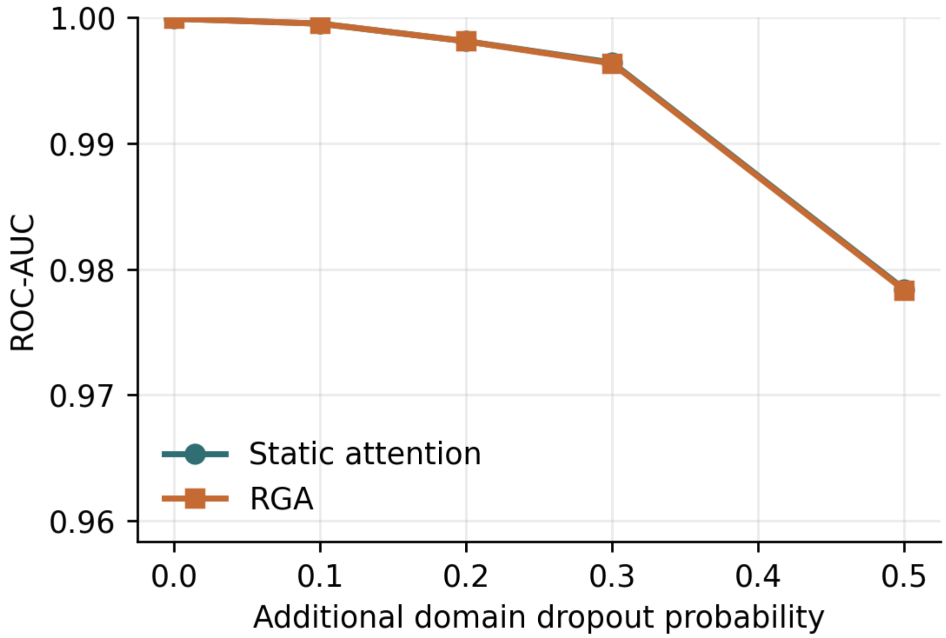

Figure 3: ROC-AUC under additional missing-domain dropout.

5.3 Score Drift

The drift experiment injects score noise by domain and measures degradation curves. Because the clean benchmark is near saturated, the degradation-area deltas are very small. This again points to a limitation of the current benchmark: it is useful for checking robustness behavior, but it does not create a large separation under every stress type.

Table 8: Drift robustness summary. Higher area is better.

Domain	**Static**	**RGA**	**Delta**	**Seeds**
behavior	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	0.0000	5
cyber	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	0.0000	5
fraud	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	-0.0000	5
nlp	0.2999 [0.2999, 0.2999]	0.2999 [0.2999, 0.2999]	-0.0000	5

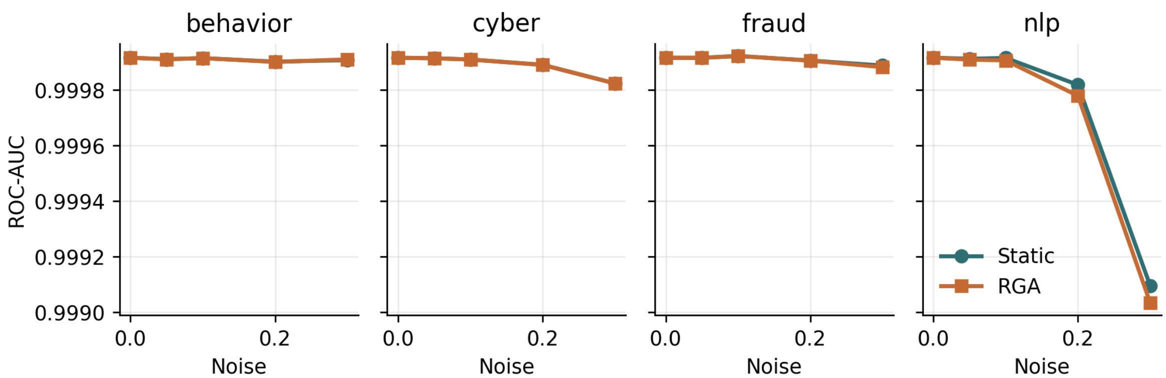

Figure 4: Per-domain score-drift curves on the real-domain benchmark.

5.4 Adversarial Score Perturbations

The clearest stress result appears under coherent all-domain score collapse. Under the all-domain zero attack, static attention obtains ROC-AUC 0.7212 while RGA obtains 0.7718 (delta +0.0506). Under the all-domain max attack, static attention obtains 0.7401 while RGA obtains 0.7720 (delta +0.0319). Under all-domain Gaussian noise, RGA is essentially tied with static attention (delta −0.0001). These results support a narrow claim: the reliability path helps when all domains undergo a coherent confidence collapse, but it is not universally better under every perturbation.

Table 9: Adversarial perturbation benchmark. Delta is RGA minus static attention.

Attack	Target	Static ROC	RGA ROC	Delta	Delta 95% CI
gaussian_noise	all	0.9999	0.9998	-0.0001	[-0.0001, 0.0000]
gaussian_noise	behavior	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]
gaussian_noise	cyber	0.9999	0.9999	0.0000	[0.0000, 0.0000]
gaussian_noise	fraud	0.9999	0.9999	0.0000	[0.0000, 0.0000]
gaussian_noise	nlp	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]
max_attack	all	0.7401	0.7720	0.0319	[0.0050, 0.0617]
max_attack	behavior	0.9997	0.9997	-0.0000	[-0.0000, -0.0000]
max_attack	cyber	0.9972	0.9972	0.0000	[0.0000, 0.0000]
max_attack	fraud	0.9996	0.9996	0.0000	[0.0000, 0.0000]
max_attack	nlp	0.9840	0.9843	0.0003	[-0.0009, 0.0016]
zero_attack	all	0.7212	0.7718	0.0506	[0.0315, 0.0681]
zero_attack	behavior	0.9997	0.9997	0.0000	[-0.0000, 0.0000]
zero_attack	cyber	0.9980	0.9980	0.0000	[0.0000, 0.0000]
zero_attack	fraud	0.9995	0.9995	0.0000	[0.0000, 0.0000]
zero_attack	nlp	0.9713	0.9716	0.0003	[-0.0004, 0.0009]

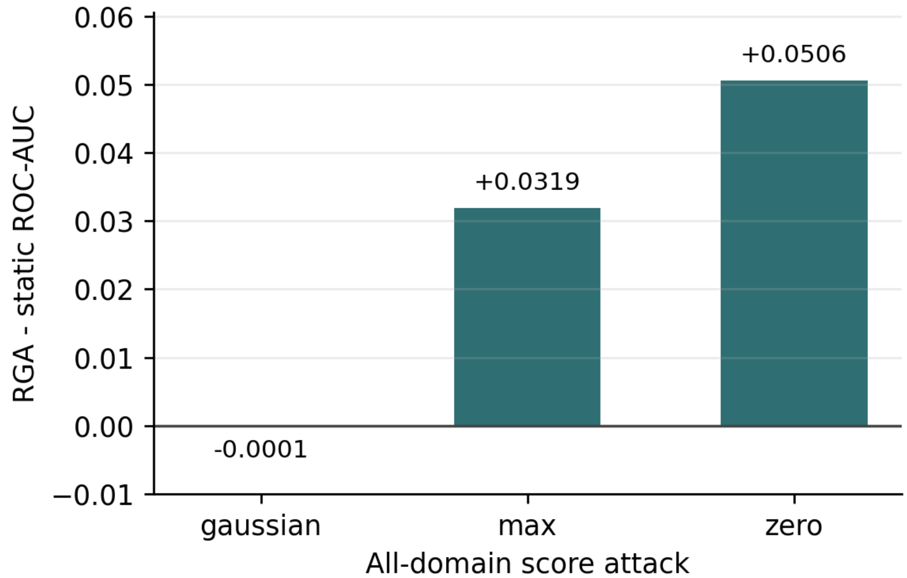

Figure 5: All-domain adversarial ROC-AUC delta between RGA and static attention.

6 Ablation and Failure Analysis

6.1 Reliability-Gate Threshold Sweep

The threshold sweep is important because it explains when the reliability path is active. Table 10 reports results for $\tau \in \{0.4, 0.5, 0.6, 0.66, 0.7, 0.8, 0.9\}$. At the configured $\tau = 0.66$, the clean adaptation rate is 0.032. This means that, on clean data, almost all samples use the static path. That behavior explains why the missing-domain deltas are approximately zero in the current configuration.

Under all-domain zero and max score collapse, the same threshold activates the reliability path for every evaluated sample. This is the main mechanism behind the stress gains. Lower thresholds through $\tau = 0.60$ suppress adaptation and lose the gains. Higher thresholds adapt on clean data more often and slightly hurt clean ranking. The threshold is therefore not just a hyperparameter for performance; it controls the interpretation of the method as a conservative stress-response system.

Table 10: Reliability-gate threshold sweep. Delta is RGA minus static attention; adaptation rate is the fraction of evaluated samples using the reliability-aware path.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.50	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.60	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.66	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.70	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]	0.320
clean	0.80	0.9999	0.9999	-0.0000	[-0.0001, 0.0000]	0.960
clean	0.90	0.9999	0.9999	-0.0000	[-0.0001, 0.0000]	1.000
gaussian_noise:all	0.40	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.50	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.60	0.9999	0.9998	-0.0000	[-0.0001, -0.0000]	0.680
gaussian_noise:all	0.66	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.70	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.80	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.90	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
max_attack:all	0.40	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.60	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.66	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.70	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.80	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.90	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
zero_attack:all	0.40	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.60	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.66	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.70	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.80	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.90	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000

6.2 Reliability-Component Ablation

The component ablation gives one of the most useful findings in the chapter. Removing the ECE term does not harm the all-domain collapse cases. In fact, the no-ECE variant raises the max-attack delta from 0.0319 to 0.0457 and the zero-attack delta from 0.0506 to 0.0629. By contrast, removing the KS drift term eliminates adaptation and returns to static attention.

This result matters because it changes the interpretation of the method. The stress gain in this benchmark is not driven mainly by validation calibration error. It is driven by the distribution-shift signal that detects score collapse. ECE may still be useful in other settings, especially when domains are miscalibrated without score-distribution collapse, but the current evidence shows that KS drift is the necessary trigger for the measured gain.

Table 11: Reliability-component ablation on all-domain score attacks. Delta is RGA minus static attention.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
full	max_attack	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
full	zero_attack	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
no_ece	gaussian_noise	0.9999	0.9996	-0.0003	[-0.0005, -0.0001]	1.000
no_ece	max_attack	0.7401	0.7858	0.0457	[0.0129, 0.1067]	1.000
no_ece	zero_attack	0.7212	0.7841	0.0629	[0.0493, 0.0808]	1.000
no_gate	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
no_gate	max_attack	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
no_gate	zero_attack	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
no_ks	gaussian_noise	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
no_ks	max_attack	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
no_ks	zero_attack	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
no_sharpness	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
no_sharpness	max_attack	0.7401	0.7808	0.0408	[0.0123, 0.0811]	1.000
no_sharpness	zero_attack	0.7212	0.7759	0.0547	[0.0382, 0.0716]	1.000

6.3 Cross-Benchmark Contrast (MVTec 3D-AD vs Label-Aligned)

The same mechanism-isolation protocol applied to the naturally paired MVTec 3D-AD benchmark produces the opposite verdict. On MVTec 3D-AD, the KS-drift term is again the only component that controls whether the gate fires – removing it (the `no_ks` variant) returns adaptation to 0% and deltas to zero. But every condition that does fire the gate produces a *negative* ROC-AUC delta of approximately -0.10 to -0.13 relative to static attention, with 95% confidence intervals that exclude zero. The `no_ece` variant on MVTec 3D-AD is marginally better than the full configuration on some attacks, mirroring the label-aligned finding, but the absolute delta remains strongly negative in every case.

Table 12: MVTec 3D-AD reliability-component ablation on all-domain score attacks.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.6495	0.5188	-0.1307	[-0.1870, -0.0825]	1.000
full	max_attack	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
full	zero_attack	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000
no_ece	gaussian_noise	0.6495	0.5261	-0.1234	[-0.1877, -0.0809]	1.000
no_ece	max_attack	0.6330	0.5315	-0.1015	[-0.1612, -0.0402]	1.000
no_ece	zero_attack	0.6442	0.5299	-0.1143	[-0.1612, -0.0704]	1.000
no_gate	gaussian_noise	0.6495	0.5188	-0.1307	[-0.1870, -0.0825]	1.000
no_gate	max_attack	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
no_gate	zero_attack	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000
no_ks	gaussian_noise	0.6495	0.6495	0.0000	[0.0000, 0.0000]	0.000
no_ks	max_attack	0.6330	0.6330	0.0000	[0.0000, 0.0000]	0.000
no_ks	zero_attack	0.6442	0.6442	0.0000	[0.0000, 0.0000]	0.000
no_sharpness	gaussian_noise	0.6495	0.5314	-0.1181	[-0.1633, -0.0796]	0.921
no_sharpness	max_attack	0.6330	0.5180	-0.1150	[-0.1803, -0.0383]	1.000
no_sharpness	zero_attack	0.6442	0.5188	-0.1253	[-0.1822, -0.0775]	1.000

Table 13: MVTec 3D-AD reliability-gate threshold sweep. Delta is RGA minus static attention; adaptation rate is the fraction of evaluated samples using the reliability-aware path.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.6497	0.6497	0.0000	[0.0000, 0.0000]	0.000
clean	0.50	0.6497	0.6497	0.0000	[0.0000, 0.0000]	0.000
clean	0.60	0.6497	0.6497	0.0000	[0.0000, 0.0000]	0.000
clean	0.66	0.6497	0.6098	-0.0398	[-0.1206, 0.0000]	0.204
clean	0.70	0.6497	0.5812	-0.0684	[-0.1226, -0.0063]	0.404
clean	0.80	0.6497	0.5317	-0.1180	[-0.1506, -0.0901]	0.879
clean	0.90	0.6497	0.5225	-0.1271	[-0.1757, -0.0845]	1.000
gaussian_noise:all	0.40	0.6502	0.6502	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.50	0.6502	0.6502	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.60	0.6502	0.5478	-0.1025	[-0.1380, -0.0698]	0.800
gaussian_noise:all	0.66	0.6502	0.5267	-0.1235	[-0.1882, -0.0576]	1.000
gaussian_noise:all	0.70	0.6502	0.5267	-0.1235	[-0.1882, -0.0576]	1.000
gaussian_noise:all	0.80	0.6502	0.5267	-0.1235	[-0.1882, -0.0576]	1.000
gaussian_noise:all	0.90	0.6502	0.5267	-0.1235	[-0.1882, -0.0576]	1.000
max_attack:all	0.40	0.6330	0.6330	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.6330	0.6330	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.60	0.6330	0.6330	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.66	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
max_attack:all	0.70	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
max_attack:all	0.80	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
max_attack:all	0.90	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
zero_attack:all	0.40	0.6442	0.6442	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.6442	0.6442	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.60	0.6442	0.6442	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.66	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000
zero_attack:all	0.70	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000
zero_attack:all	0.80	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000
zero_attack:all	0.90	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000

The asymmetry between the two benchmarks is the central scientific finding of this chapter. The KS-drift signal that delivers the entire +0.0506 `zero_attack` gain on label-aligned synthetic pairing *mis-fires* on legitimate inter-category heterogeneity in naturally paired data, places the mean reliability below τ on most batches, and the resulting adapted weighting consistently underperforms the static attention solution. The cleanest reading of the two results together is that a validation-derived drift detector cannot tell adversarial coherent score collapse apart from natural cross-category variation on this scale of paired data. The mechanism therefore needs a benchmark-conditioned drift threshold, a different drift signal, or a learned gate before it can be deployed on natural paired evidence streams.

6.4 Calibration and Domain Attribution

Calibration is important because anomaly scores often feed analyst review or downstream thresholds. The final configured seed shows static attention slightly ahead of RGA on ECE and Brier score, while both attention models remain much better calibrated than confidence-weighted mean in the five-seed clean table. This is another reason the chapter avoids a blanket superiority claim.

Table 14: Calibration and counterfactual domain-attribution summary.

Metric	**Static**	**RGA**
ECE	0.0038	0.0038
Brier	0.0032	0.0032
CDA samples	–	400
CDA/ECE Spearman	–	0.052
CDA/ECE Spearman status	–	computed

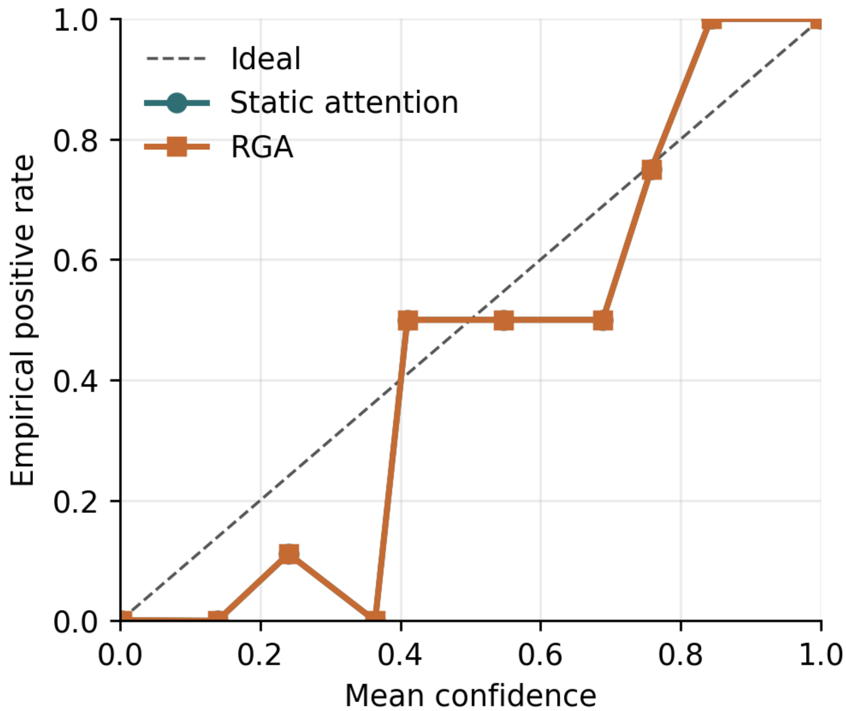

Figure 6: Reliability diagram for static attention and RGA.

Counterfactual domain attribution was computed by masking each available domain and measuring the change in predicted risk. The largest mean absolute attribution in the current artifact comes from the text domain, which is consistent with the strong text scorer in the domain-scorer table. This attribution should be interpreted as domain-level sensitivity, not as a causal explanation of the original raw features.

Figure 7: Mean absolute counterfactual domain-attribution impact.

6.5 Failure Cases

Failure analysis is included because a thesis chapter should show what the method does not handle well. The result artifact exports representative cases: the largest RGA win, the largest RGA loss, cases where RGA fixes static attention, cases where static attention fixes RGA, and high-confidence RGA failures. Each case includes the sample identifier, label, static and RGA probabilities, per-domain scores, missing-domain mask, and reliability weights.

Table 15: Representative failure and contrast cases from the latest configured seed.

Case	**Label**	**Static** p	**RGA** p	**Sample**
biggest_elara_win	0	0.0001	0.0001	real_fusion_test_000000
biggest_elara_loss	1	0.9998	0.9998	real_fusion_test_001072
high_confidence_elara_failure	1	0.0470	0.0470	real_fusion_test_001517

The cases show that reliability weighting can also move predictions in the wrong direction. This is expected for a gate based on aggregate reliability signals. It does not inspect the full semantic content of each domain. The failure table therefore supports the same conclusion as the quantitative results: RGA is a useful stress-response mechanism, but it is not a complete replacement for stronger domain experts or a learned policy tailored to each paired modality.

7 Threats to Validity

7.1 Internal Validity

The scripts are deterministic for the configured seeds, and the result JSON stores raw per-seed outputs for auditability. The main internal-validity risk is that the stress tests are synthetic. Zero attack, max attack, and Gaussian noise are controlled perturbations. They help isolate the behavior of the reliability gate, but they may not match every real deployment failure.

7.2 Construct Validity

The largest construct-validity issue is benchmark construction. ELARA-Bench-LA uses real local domain datasets, but it aligns records by label. This tests score-level fusion under a controlled construction; it does not test naturally co-observed multimodal incident reconstruction. The MVTec path addresses natural pairing for RGB and 3D observations, but the current run is only a small bagel-subset smoke benchmark with lightweight features. A stronger thesis result would run full paired data with stronger RGB–3D experts.

7.3 External Validity

The results should not be interpreted as deployment evidence. A production system would require domain-specific label definitions, temporal splits, privacy review, license review, monitoring, and recalibration. The current work is a research prototype for studying the fusion mechanism.

7.4 Statistical Validity

Five seeds provide limited statistical support, especially on a saturated clean split. Confidence intervals are reported to make this visible. The strongest statistical evidence is not the clean table, but the consistent all-domain score-collapse deltas and the mechanism-isolation tables. Even there, the chapter treats the result as conditional because the stress setup is controlled.

8 Reproducibility

The repository supports a scripted reproduction flow:

1. prepare the real-domain fusion inputs;
2. optionally download and prepare the paired MVTec 3D-AD subset;
3. run the fusion benchmark YAML configurations;
4. generate tables and figures from the result JSON artifacts;
5. compile the manuscript with Tectonic.

The relevant entry points are the benchmark-preparation scripts under `src/scripts/`, the fusion configurations under `configs/`, the generated result artifacts under `experiments/fusion/`, and the generated tables and figures under `docs/research/`. Regression tests cover attention masking, reliability estimation, statistics, baseline inclusion, paired-benchmark metadata, float XYZ TIFF reading, and result aggregation.

9 Conclusion and Future Work

This chapter presented ELARA, a reliability-aware score-fusion research system, and RGA, a conservative reliability-gated attention module. The main result is narrow but useful. On a saturated label-aligned benchmark, RGA preserves static attention performance. Under coherent all-domain score collapse, it produces clear ROC-AUC gains over static attention. The threshold sweep shows why this happens: at $\tau = 0.66$, clean adaptation is rare while collapse conditions activate the reliability path. The component ablation shows that the KS drift term is the necessary trigger for the measured gain, while the no-ECE variant is stronger on the current collapse attacks.

The chapter also reports a stronger negative result that did not exist in earlier drafts. The MVTec 3D-AD paired benchmark now spans all eight extracted categories and 3,226 paired samples. On that data, random forest fusion is the strongest method by a clear margin, static attention is competitive but mid-pack, and the reliability gate actively lowers ROC-AUC on every condition including clean. The cross-benchmark contrast in §6.3 isolates the cause: the KS-drift signal that delivers the entire label-aligned stress gain mis-fires on legitimate inter-category variation in naturally paired data. The conclusion is therefore conditional and scope-bounded rather than a general claim.

The next research steps are clear. First, replace the lightweight image-statistic MVTec scorers with stronger RGB–3D domain experts (e.g. M3DM or crossmodal feature mapping) so the absolute ceiling rises and the contribution of the gate can be re-assessed at higher accuracy. Second, run the reliability gate against additional naturally co-observed anomaly datasets to see whether the misfire pattern is specific to MVTec 3D-AD's category heterogeneity or general to natural-paired data. Third, replace the fixed threshold with a learned gate (already prototyped in `src/uais/fusion/attention/learned_gate.py`) and evaluate whether data-driven gating closes the natural-paired regression. Fourth, develop a formal switching-rule analysis that explains when reliability-aware fusion should be preferred to the static path under bounded drift assumptions.

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